Introductory Lecture

Gender inequality and the labour market
October 2, 2024

Gender gaps

The gender wage gap



Are gender wage gaps evidence of discrimination?

- It could be that men and women have different levels of **human capital**
- Blinder and Oaxaca explored whether men and women are paid differently for the same human capital characteristics by running gender-specific wage regressions

$$\ln w_{gi} = x_{gi}\beta_g + \varepsilon_{gi}$$

- where $g \in m, f$ denotes gender and x_{gi} contains individual i's human capital characteristics and a gender-specific constant

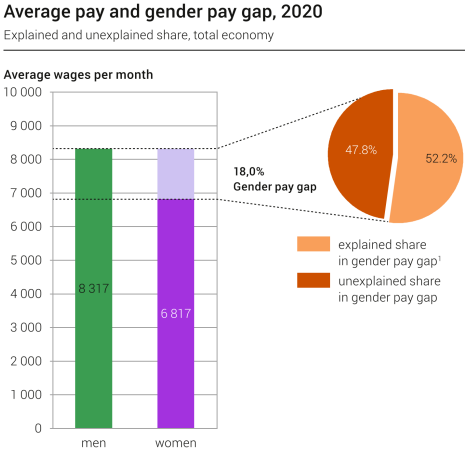
Blinder-Oaxaca decomposition

- The gender wage gap can be deconstructed using the gender-specific estimates of the coefficient vectors $\hat{\beta}_m$ and $\hat{\beta}_f$, the gender-specific average log wages $\ln \bar{w}_m$ and $\ln \bar{w}_f$, and the gender-specific average characteristics $\bar{x}_m$ and $\bar{x}_f$

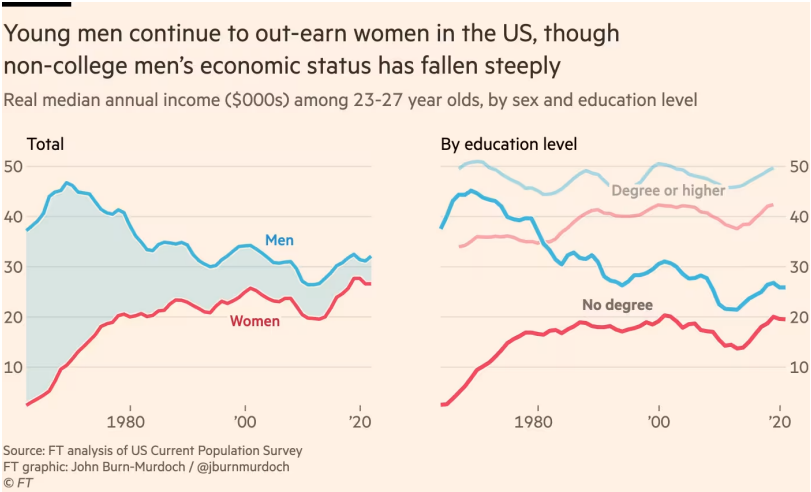
$$\ln \bar{w}_m - \ln \bar{w}_f = (\bar{x}_m - \bar{x}_f)\hat{\beta}_m + \bar{x}_f(\hat{\beta}_m - \hat{\beta}_f)$$

- The first term represents the **"explained"** part of the wage gap
- The second term captures the **"unexplained"** part of the wage gap
- Oaxaca (1973) found that 42% of the wage gap was explained by variables including education, industry, employment type, location
- Concluded that 58% of the gap was due to **discrimination**

Explained and unexplained gender wage gap in Switzerland

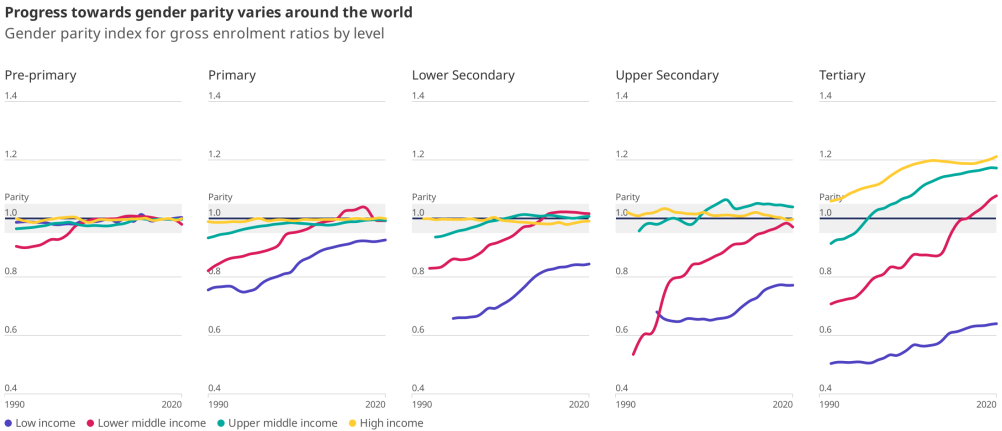


Changes in the gender wage gap



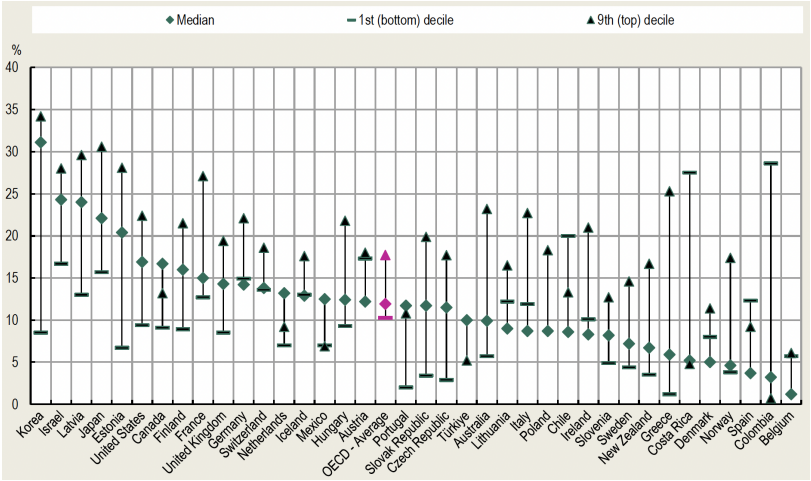
Source: Financial Times 2024.

Gender gaps in education



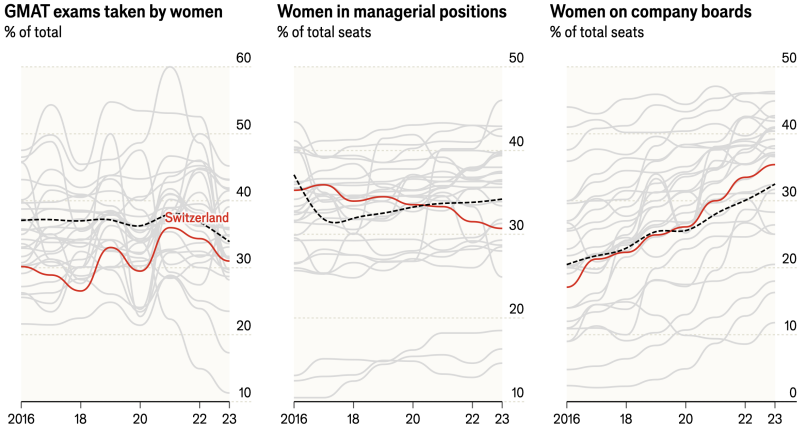
This graphic was designed for the GEM Report SCOPE website. www.education-progress.org
Selection: income.
Source: UIS database, 2022. Accessed: Sep 27, 2024

The gender wage gap across the earnings distribution



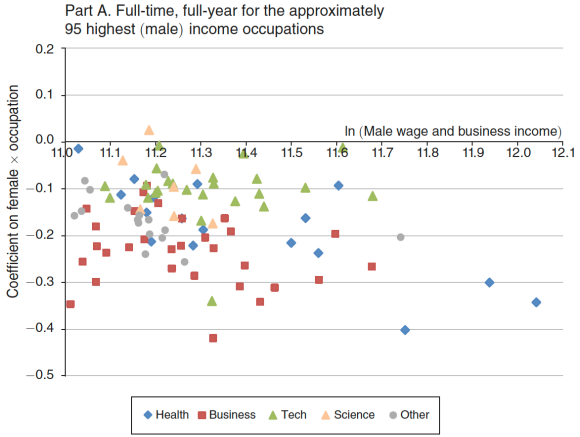
Notes: The graph shows the gender wage gap at the 1st decile, at the median, and at the 9th decile, for full-time employees, 2020 or latest year available. The gender wage gap is unadjusted and is calculated as the difference between earnings of men and women, relative to men. Source: OECD 2023.

Glass ceilings



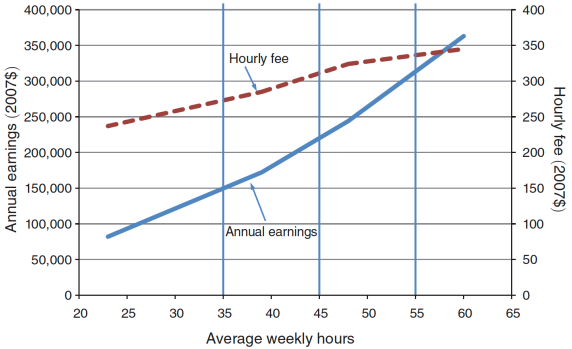
Notes: The dashed line shows the OECD average. The red line shows Switzerland. Source: The Economist 2024.

Gender wage gaps by occupation



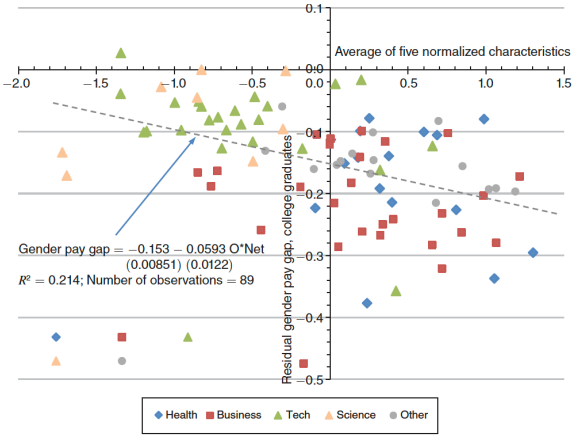
Notes: The figure shows the gender wage gap and male wages for the 95 highest income occupations. Source: Goldin 2014.

Nonlinear earnings



Notes: The figure shows hourly fees and annual earnings for lawyers. Source: Goldin 2014.

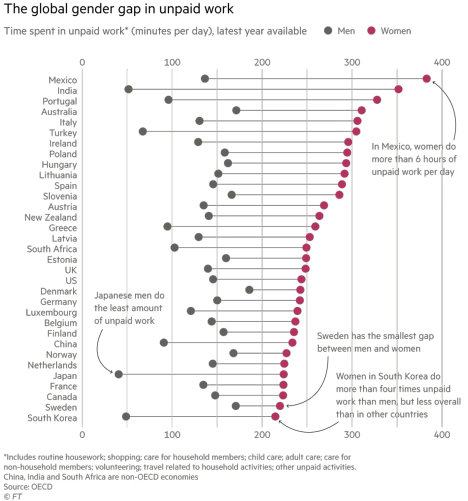
Inflexibility and gender wage gaps



Notes: The figure shows the relationship between the average of 5 normalized O*Net characteristics and the gender wage gap. The O*Net characteristics reflect different aspects of job inflexibility (time pressure, contact with others, relationships, structured work, decision-making freedom). Source: Goldin 2014.

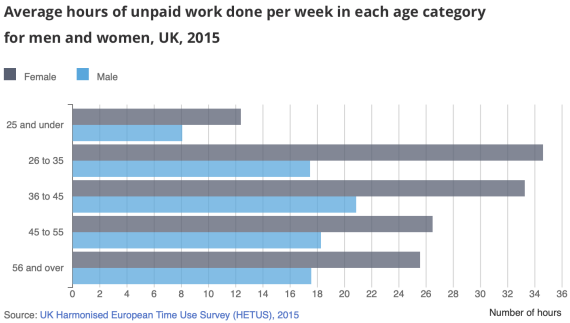
Household economics

Unequal division of housework



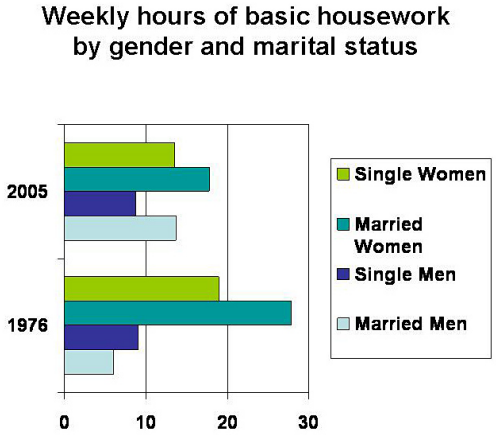
Source: Financial Times 2019

Unequal division of housework across age groups



Source: ONS 2016

Improvements over time



Source: U-M Institute for Social Research (ISR) Panel Study of Income Dynamics

Housework and the gender wage gap

- Becker 1985 developed a model in which the unequal division of housework is a **rational** decision on the part of the household
- Unequal housework division can explain the gender wage gap and occupational gender segregation
- In the model, individual's earnings from a market activity are proportional to the time spent on the activity and the stock of human capital specific to that activity

Gains from specialization and trade

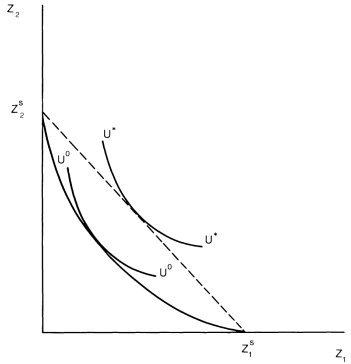


FIG. 1.—The gains from specialization and trade

- Under certain assumptions, individuals specialize

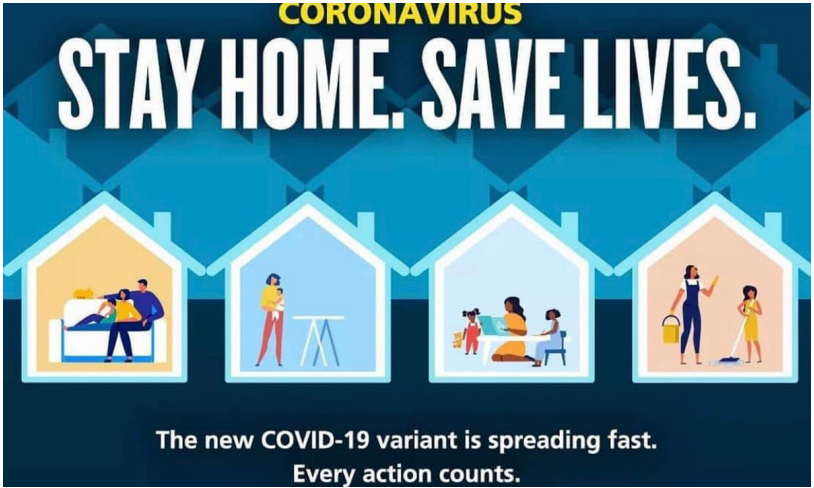
Specialization in the household

- Housework and market activities both require **effort**
- Individuals in a household are effort-constrained
  - Increased effort on housework reduces effort on market activities, which reduces productivity and thus wages
- Specialization is optimal:
  - One partner should do the housework to enable the other partner to spend invest more effort in market activities
  - Specialization increases household earnings
- But this does not predict that **women** specialize in housework...

Comparative advantage and outcomes

- Women's comparative advantage in housework may arise from **discrimination**
- Even small discriminatory differences in hourly pay may lead to large differences in earnings
  - Because **comparative advantage** leads to gender differences in human capital accumulation
  - These differences in human capital would not have arisen were it not for discrimination
- The model can also explain **occupational segregation**
  - Because women seek occupations that are more compatible with housework
- The model predicts that single women have higher hourly pay than married women

COVID-19 and the division of housework



Source: The Guardian 2021

COVID-19 and the division of housework

- COVID-19 lockdowns had several impacts on household economics
  - Job losses, particularly in sectors with frequent interactions between customers and providers
  - Increases in home production, due to school and kindergarten closures
- How did COVID-19 affect gender inequalities?
  - Hupkau and Petrongolo 2020 find that women provided 60% of the increased childcare needs
  - Hupkau and Petrongolo 2020 also find that in 20% of households, men became the main childcare provider
  - Adams-Prassl et al. 2020 find that women were more likely to lose their jobs due to increased childcare responsibilities

Child penalty

Why do we care about the child penalty?

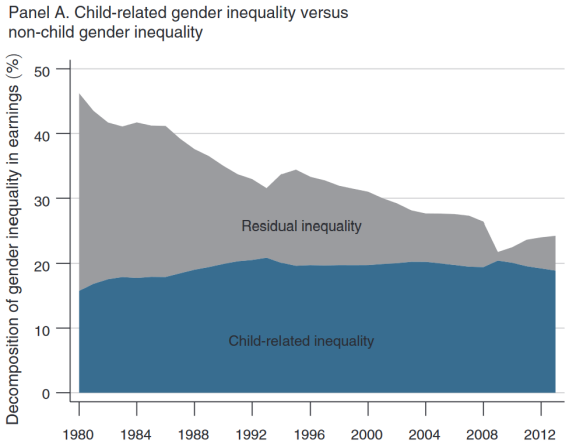


Figure: Taken from Kleven, Landais, and Søgaard 2019, Fig. 5.

What is the child penalty?

- Treatment D: Woman has a child
- Potential outcome $Y_{i,d}$: outcome of individual i when $D = d$
- Potential outcomes at different time horizons: $Y_{i,d,\mathbf{t}}$ (t: time since treatment)
- Problem statement: Figure out $E\left[Y_{i,1,t} - Y_{i,0,t} | D = 1\right]$, $t \geq 0$, or something close to it

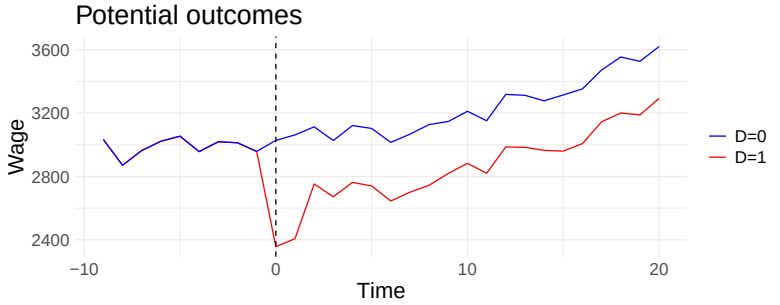


Illustration of Potential outcomes and individual treatment effect $Y_{i,1,t} - Y_{i,0,t}$ for a specific individual i.

How do we estimate the child penalty?

How do we figure out $E[Y_{i,d=0,t}|D=1]$? $\Rightarrow$ Pick a control group!

- Match with other women based on pre-birth characteristics (Kleven, Landais, and Søgaard 2019; Kleven et al. 2019)
  - Long-run penalty 20-50% relative to control group
  - Some recent criticisms of this approach: Melentyeva and Riedel, n.d.; Bensnes, Huitfeldt, and Leuven 2023
- IV approaches:
  - IVF treatment (Lundborg, Plug, and Rasmussen 2024): Long-run child **premium**!
  - Sibling sex mix (Aaronson et al. 2021)
- ...

Kleven

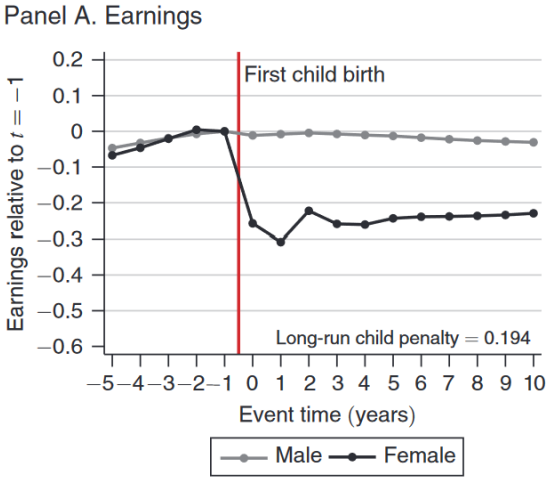


Figure: Taken from Kleven, Landais, and Søgaard 2019

How does the child penalty arise?

- Kleven et al. 2019: decompose child penalty into three components
  - Extensive margin (participation). Important in UK & US
  - Earnings: hours worked $\times$ wage rate (can't distinguish in many administrative datasets). Important in Scandinavia & German-speaking countries
  - Kleven, Landais, and Søgaard 2019: Mothers less likely to obtain leadership positions; favour firms offering amenities (esp. family-friendly) over pecuniary rewards
- Possible reasons (interactions possible):
  - Short- & long-term health shock (Andresen and Nix 2022)
  - Preferences & social norms (see below)
  - Convex returns (Goldin 2014)
  - Lack of (formal & informal) childcare (Olivetti and Petrongolo 2017; Kleven et al. 2024).
  - Monetary reasons (Olivetti and Petrongolo 2017)
  - Non-wage factors (amenities) (Wasserman 2023)
  - Discrimination (see below)
  - Misinformation (Costa-Ramon et al. 2024)
- Grandmother penalty

Participation in Switzerland

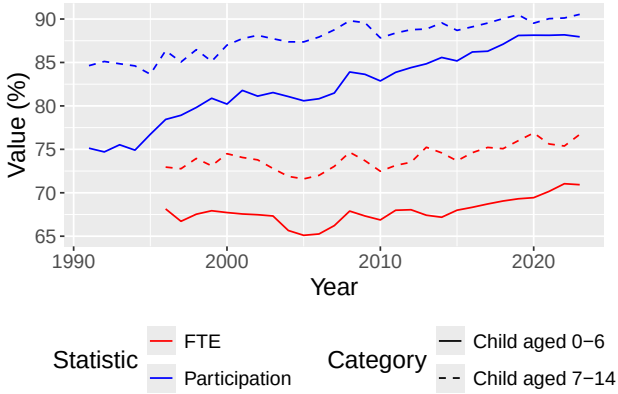


Figure: Labour market participation of women in Switzerland, aged 15-64 over time. Source: own illustration; composite statistics from BFS, Sektion Arbeit und Erwerbsleben, Erwerbsquoten nach Geschlecht, Nationalität, Altersgruppen, Familientyp & Erwerbsquoten in Vollzeitäquivalenten nach Geschlecht, Nationalität, Altersgruppen, Familientyp, both released 20 Aug 2024; based on SAKE. Downloaded 2 Sep 2024.

Gender norms & stereotypes

Why care about norms?

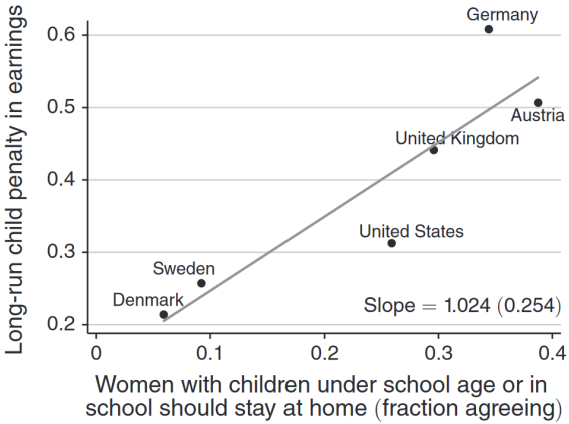


Figure: Fig. 4 of Kleven et al. 2019: Estimated child penalty vs. elicited gender norms based on cross-section of countries

A society-wide problem?

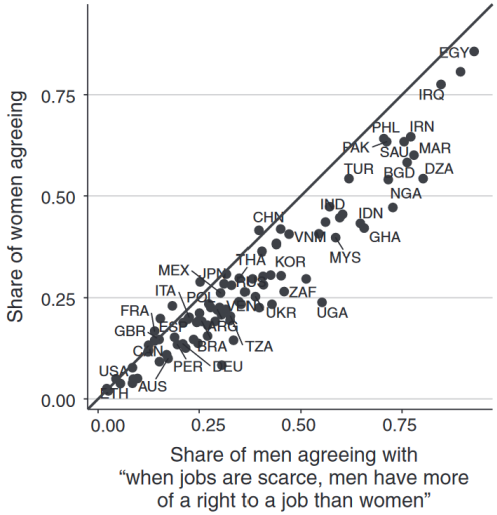


Fig. 4 of Bertrand 2020: Share of men vs. share of women agreeing with the statement: *"When jobs are scarce, men have more right to a job than women."*
Based on a cross-section of countries, data from IVS collected in the 2000s.

A society-wide problem?

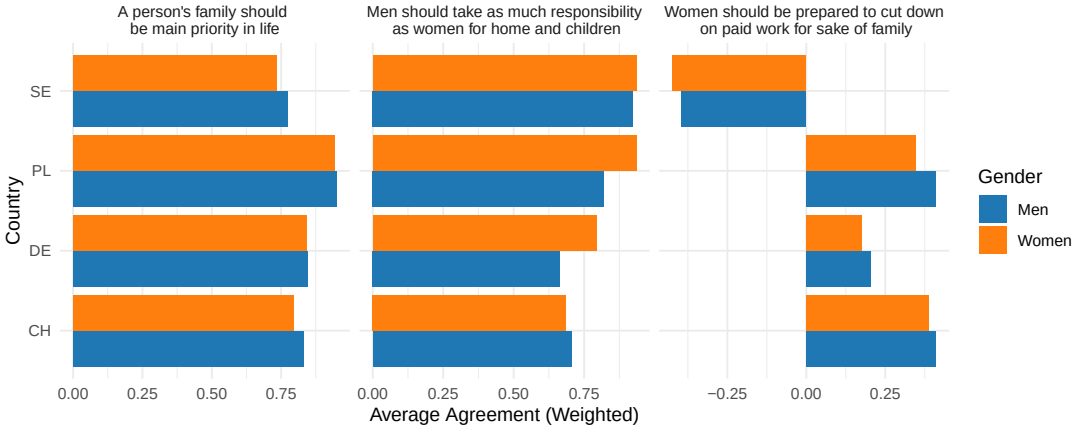


Figure: Questions about social norms by gender. Calculations and graph own, based on survey data from ESS. PSPweight applied. If multiple waves are available, pools waves

Is there progress?

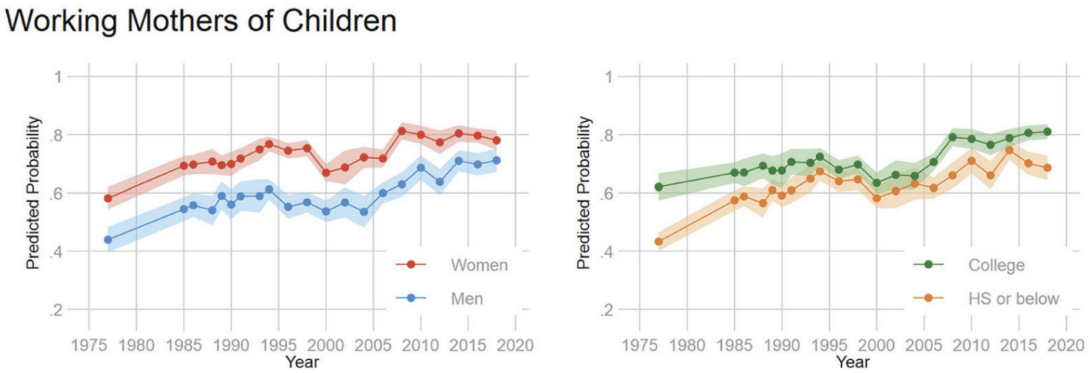


Figure: Fig. 1 of Meagher and Shu 2019: Prediction of egalitarian attitudes towards working mothers over time in the US. Based on GSS data

How do preferences & norms affect gender pay gap?

- Gender gaps in occupational choice, risk-taking affected by norms (Gneezy, Leonard, and List 2009)
- Social norms lead to worse outcomes & self-sabotage of women (Bursztyn, Fujiwara, and Pallais 2017; Bertrand, Kamenica, and Pan 2015)
- Womens' and couples' choices are affected by perception of social norms (Bursztyn, González, and Yanagizawa-Drott 2020)
- Non-traditional partnerships less affected by child penalty (Andresen and Nix 2022)
- Child penalty inherited matrilineally (Kleven, Landais, and Søgaard 2019)

Risk-taking in a matrilineal society

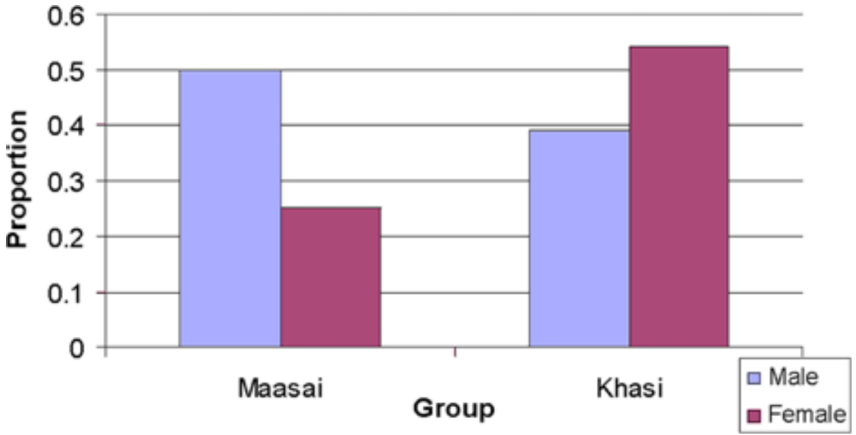


Figure: Fig. 1 of Gneezy, Leonard, and List 2009: A summary of [pro-]competitive choices across gender in the two societies. The Maasai are a patrilineal, the Khasi a matrilineal society.

Social desirability

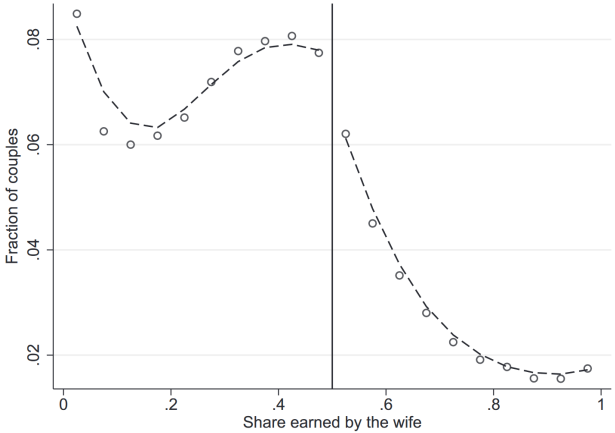


Figure: Fig. 1 of Bertrand, Kamenica, and Pan 2015 Distribution of relative income (married individuals between 18 and 65 years of age & where both husband & wife earn positive income.

Discrimination

Detour: How economists used to think about discrimination

Becker's framework:

- Taste-based discrimination
  - Employer hires green: $u^{green} = \pi - w$
  - Employer hires blue : $u^{blue} = \pi - w - x$
  - Consequence: Employer hires blue only at a lower wage
  - Consequence: Perfect markets; customer doesn't discriminate & some firms don't discriminate $\rightarrow$ no discrimination (employer needs to "pay" for right to discriminate)
- Statistical Discrimination
  - Suppose $a^c \sim N(\mu^c, \sigma^2)$, $\mu^{green} > \mu^{blue}$; profits are multiplied with a^c $(E[u^c] = E[a^c]\pi - w)$
  - Employer hires green: $E[u^{green}] = \mu^{green}\pi - w$
  - Employers hires blue: $E[u^{blue}] = \mu^{blue}\pi - w$
  - Consequence: Employer hires blue only at a lower wage
  - Consequence: Discrimination is profit-maximizing $\rightarrow$ Doesn't vanish with perfect competition (perfect competition $\rightarrow$ firm goes bankrupt if it **doesn't** discriminate.
  - can even cause self-fulfilling prophecy! (Coate and Loury 1993)

A better way to think about discrimination?

Becker: Discrimination is performed consciously & with intent
Alternatively, Implicit Discrimination (Bertrand, Chugh, and Mullainathan 2005):
"Unconscious mental associations between a target [...] and a given attribute" (Try out the Implicit Attitudes test)

- Differential behaviour a result of subconscious processes in brain
- Spills over into work esp. when conditions are "messy, pressured, and distracting"
- Can be influenced: more conscious choice; manipulation

c.f. stereotyping

Where is discrimination today? - hiring

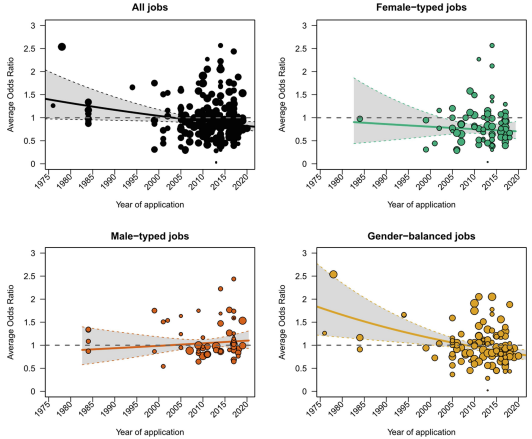


Figure: Fig. 4 from Schaerer et al. 2023: Gender preference in job hirings by type of job. Odds-ratios $>$ 1 indicate preference for men, odds-ratios $<$ 1 indicate preference for women. Results based on meta-analysis.

Where is discrimination today - promotion

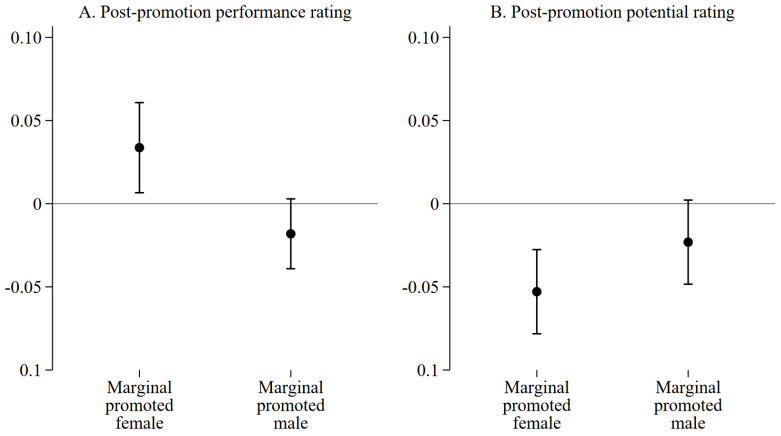


Figure: Fig. 5 of Benson, Li, and Shue 2024 Performance and potential rating for men & women. Two suggested channels: Differential attrition & sources of attrition; Gender attitudes.

Gender-based violence

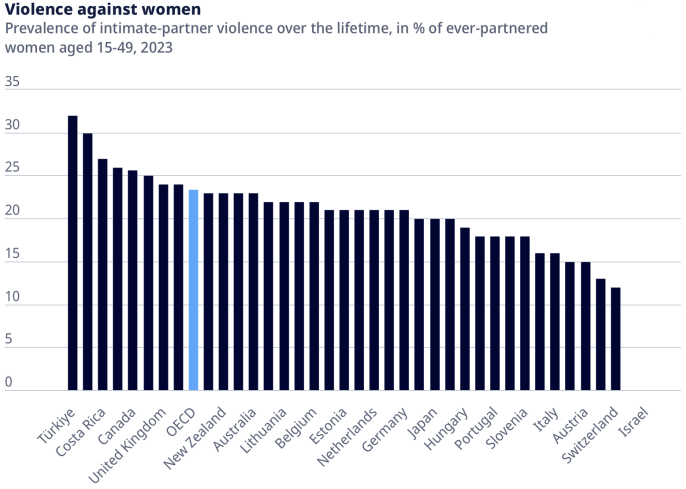
Gender-based violence

- **Gender-based violence:** harmful acts directed against a person because of their gender or violence that disproportionately affects people of a certain gender
  - Harmful acts can be physical, sexual, mental or economic
  - Can be inflicted in public or in private
  - Often goes unreported
- Focus on two forms of gender-based violence:
  1. **Domestic violence**
  2. **Violence in the workplace**

Why is gender-based violence an economic issue?

1. Economic costs
   - Loss of earnings, health services, social welfare, criminal justice system services, physical/emotional impacts
   - In the EU, gender-based violence estimated to cost EUR 366 billion per year (EIGC 2021)
   - In Vietnam, violence against women costs around 1.4% of GDP (UN Women 2012)
2. Obstacle to gender equality in the labour market
   - Negative impacts of domestic violence on women's employment and earnings (Adams et al. 2024)
   - Workplace violence increases gender inequality in the labor market (Adams-Prassl et al. 2024, Folke and Rickne 2022)
3. Economic causes
   - Decreases in the wage gap decrease incidents of domestic violence (Aizer 2010)
   - Decrease in unemployment benefits increases underreporting of workplace sexual harassment (Dahl and Knepper 2021)

Domestic violence



Notes: The figure shows the percentage of women who have experienced physical and/or sexual violence from an intimate partner at some time in their life.
Source: OECD 2023.

Aizer (2010) - The Gender Wage Gap and Domestic Violence

Theory and research question

2 theories with different predictions about the effect of the gender wage gap on DV:

1. Male backlash: female independence challenges norm, violence reinstates male authority
   - Prediction: smaller wage gap **increases** DV
2. Household bargaining model: utility function of consumption and violence, increasing women's relative wages increases bargaining power by increasing **outside option**
   - Prediction: smaller wage gap **decreases** DV

Research question: what is the impact of the gender wage gap on violence against women?

Aizer (2010) - The Gender Wage Gap and Domestic Violence

Challenges with causal identification

Two challenges with estimating the **causal** impact of the wage gap on domestic violence:

1. Endogeneity due to omitted variable bias and reverse causality
   - Omitted variable bias: other factors might be associated with wages and DV e.g. education
   - Reverse causality: DV negatively affects women's productivity
2. Underreporting, particularly nonrandom underreporting

Solutions to these two challenges:

1. Measure of relative wages constructed using industry wages and gender compositions of different industries
2. Use data on female hospitalizations for assault to measure DV

- Use data from 1990-2003 in California
  - In this period, the wage gap declined by 3.6 p.p. and (adjusted) female hospitalizations for assault declined by 36%

Aizer (2010) - The Gender Wage Gap and Domestic Violence

Results

TABLE 2—IMPACT OF WAGES ON DOMESTIC VIOLENCE—MAIN SPECIFICATION

	ln(female assaults) (1)	ln(female assaults) (2)	ln(female assaults) (3)	ln(drug admissions) (4)	ln(male assaults) (5)	ln(female assaults) (6)
Panel A. Ratio of wages						
Female/male wage	−1.469 [0.673]	−0.831 [0.313]	−0.813 [0.317]	−0.023 [0.072]	−0.257 [0.284]	0.119 [0.562]
Female/male wage × weekend						−1.15 [0.444]
Observations	984	982	982	887	982	616
R^2	0.91	0.95	0.96	0.99	0.99	0.96

Notes: Col (1) includes only minimal controls. Col (2) includes all controls apart from the lagged dep. var. Col (3) additionally includes the lagged dep. var.

- The narrowing of the wage gap between 1990 and 2003 (by 3.6 p.p.) can explain **9%** of the decline in female hospitalizations for assault over this period

Aizer (2010) - The Gender Wage Gap and Domestic Violence

Possible explanations

- Could it just be that the increase in women's relative wages meant that women spent less time at home and therefore had **fewer opportunities** to be exposed to domestic violence?
  - Investigate impact on assaults on weekdays and at weekends
  - If exposure reduction explains findings, would expect to see **larger** decrease on **weekdays**
  - Find that assaults on **weekends** explain **all** the decline in assaults
- Results consistent with **household bargaining** theory: a decrease in the wage gap increases women's bargaining power by increasing their **outside option**

Violence in the workplace

- **Workplace sexual harassment:** unwanted conduct of a sexual nature, or other conduct based on sex affecting the dignity of men and women at work
- Prevalent around the world:
  - MeToo movement highlighted magnitude of problem and lack of consequences for perpetrators
  - 1 in 15 worldwide experience sexual harassment during their working life (ILO 2022)
  - Gender differences: 8.2% of women and 5.0% of men in employment (ILO 2022)

Workplace sexual harassment and workplace sex ratios

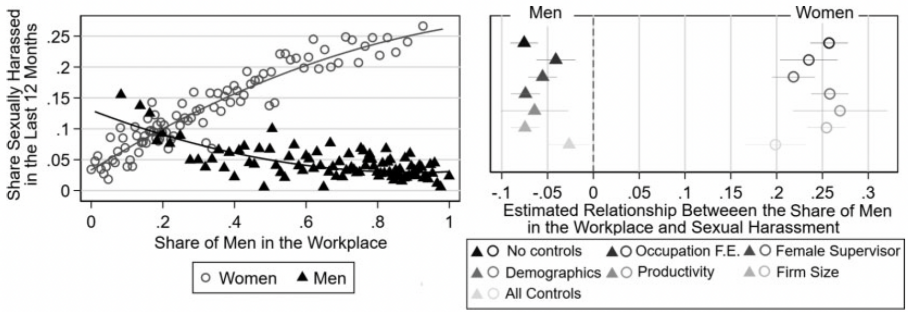


FIGURE II

Sexual Harassment Incidence across Workplace Sex Ratios

Notes: Figure 2 from Folke and Rickne 2022. The left panel shows averages for self-reports of sexual harassment from colleagues or managers. The right panel shows regression estimates from harassment on the share of men in the workplace.

Folke and Rickne (2022) - Sexual Harassment and Gender Inequality

- Conduct vignette study where ask participants to choose between fictional jobs with different characteristics
  - Including wage level and previous incidents of sexual harassment
- Find that WTP to avoid workplace where sexual harassment has occurred is equivalent to a **10% pay cut** but only when victim is of the **same** sex
- Male-dominated workplaces higher paying but entail higher disutility for women from harassment
- Through this sorting effect, sexual harassment can explain **8%** of the raw gender wage gap in Sweden

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